

Week 2: Advanced UI & JavaScript Fundamentals

3-Month MERN Stack Internship Program

This week focuses on refining frontend skills with advanced CSS techniques, building robust application-style layouts, and taking the first step into client-side scripting with JavaScript.

Day 1: Advanced CSS (Transitions, Animations, & Transforms)

- **Study:** CSS 2D/3D transforms, transition timing functions, and keyframe animations. Understand how to add visual polish without sacrificing performance.
- **Task:** Create a responsive feature showcase or team cards for a tech startup like DevPreneurs. Implement smooth hover transitions on the cards (e.g., slight scaling and box-shadow adjustments) and a subtle entry animation when the page loads.

Day 2: Forms, Accessibility (a11y), and UI Polish

- **Study:** Advanced form controls, CSS pseudo-classes for validation (:valid, :invalid), and Web Accessibility Guidelines (WCAG) basics including ARIA roles.
- **Task:** Build a comprehensive Lead Management System intake form. Ensure all inputs are fully accessible to screen readers, neatly styled, and include interactive states for focus and error styling.

Day 3: Scalable CSS & Architecture

- **Study:** CSS variables (custom properties) and methodologies for maintaining scalable code in large projects, specifically the BEM (Block Element Modifier) convention.
- **Task:** Refactor your CSS code from previous tasks using strict BEM naming conventions. Implement CSS variables to establish a consistent brand color palette (primary, secondary, accent, and neutral tones).

Day 4: Complex Layouts & CRM Dashboards

- **Study:** Combining CSS Grid and Flexbox to create nested, application-style UI architectures rather than traditional scrolling webpages.
- **Task:** Design a static, responsive CRM Dashboard layout. Include a persistent sidebar navigation, a top header profile bar, and a main content area featuring statistical metric widgets and a recent activity table. Ensure the layout collapses cleanly for mobile views.

Day 5: Introduction to JavaScript & The DOM

- **Study:** JavaScript basics: Variables (`let`, `const`), Primitive Data Types, Functions, and an introduction to the Document Object Model (DOM) selection and manipulation.
- **Task:** Attach a JavaScript file to your Day 4 CRM Dashboard. Write a script to listen for a click event on a "Menu" button to toggle the visibility of the sidebar navigation on smaller screens. Additionally, add a small script that reads the user's local time and displays a dynamic greeting (e.g., "Good Morning", "Good Afternoon").

End of Week Deliverables

- **Source Code:** Commit all daily tasks sequentially. Ensure your new JavaScript logic is properly separated into a standalone `.js` file.
- **Live Link:** Deploy the interactive CRM Dashboard (with the working JS toggle) to your hosting platform (Netlify, Vercel, or GitHub Pages).
- **Code Review:** Maintain clean console logs (remove debugging statements before pushing), properly format your HTML/CSS/JS files, and verify that there are no accessibility warnings.